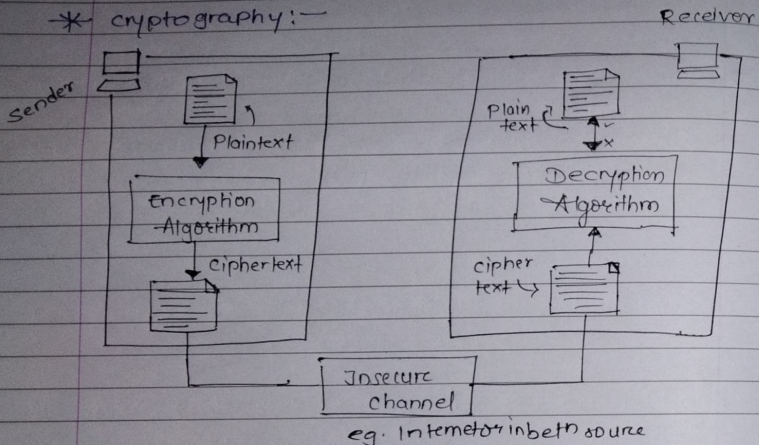


11 Feb 2026

* Answer:-

* cryptography:-



- Cryptography is an art of hiding data with the help of some technique.

* Security Goals :-

Confidentiality Integrity Availability

- Confidentiality: It is the act of protecting data from all unauthorized users.
- Integrity: It is the act of protecting the modification of data from unauthorized users.
- Availability: It is an act of protecting information or data from destruction by unauthorized users.

Confidentiality

Integrity

Availability

- | | | |
|-----------------------|-------------------------------|-------------------------------|
| 1. Protect at rest | # Hashing | # Access control |
| 2. Protect in Motion | # Access Control | # Isolation |
| 3. Protect during use | # Data classification | # Backup |
| # Encryption | # Input & Output sanitization | # Disaster Recovery |
| # Access Control | | # Business continuity process |
| # Data Classification | | |

- So, security Goals are fundamental objectives designed to protect information assets from harmful, unauthorized access or destruction.

① Confidentiality: Data should not be accessed or read without authorization. It ensures that only authorized parties have access.

② Integrity: Data should not be modified or compromised in anyway. It assumes that data remains in its intended state and can only be edited by authorized parties.

③ Availability: Data should be accessible upon legitimate request. It ensures that authorized parties have unimpeded access to data when required.

* Security Services :-

- Security services are the security services in which they are implemented into various ~~levels~~ layers and their enhancement of the security of it.
- There are the services the system provides to meet the goals.

- (a) Data Confidentiality (b) Data Integrity
 (c) Authentication (d) Non-repudiation
 (e) Access Control

(c) Authentication: Verifying the identity of the user
eg., password, faceid.

(e) Access Control: Deciding who can access what file.
eg., collg website
students only etc)

(a) Data Confidentiality: They safeguarding of data against unauthorized disclosure.

(b) Data Integrity: The guarantee of data obtained is exactly as it was submitted by legitimate source.

eg., Message from Sender to Receiver

(d) Non-repudiation: Provide security against the denial of participation in all part of conversation by an individual the communication.

* Security Mechanisms :-

There are the actual technical tools used to implement the services.

Security Service	Security Mechanism
Data confidentiality	Encipherment, Routing control
Data integrity	Encipherment, Digital signature, data integrity
Authentication	MAC, MIC, Authentication exchange
Non repudiation	Digital signature, data integrity, notarization
Access Control	Access control mechanism

- (a) Encipherment (b) Digital Signature
 (c) Traffic Padding (d) Security audit (e) Routing control

(a) Encipherment (Encryption): appln of Mechanism algorithm to transform data from the easy to difficult to understand.

(b) Digital signature: Data unit that allows a receiver to prove the data unit source.

(c) Traffic padding: Attacker sends the data instead to receiver which is false data and he can get caught when there is a gap betⁿ Sender & Receiver for long time for not sending anything which can be filled by Dummy data.

(d) Security audit: Collected data is inspected for certain reasons or acceptable, suspicious or attacked activity.

(e) Routing control: Various routes should be there so attacker could get confused.

* Relationship Security Services & Mechanisms:

- This describes how we use technical tools (Mechanisms) to achieve security goals (Services).
- One Service may require multiple Mechanisms and Mechanisms support multiple Service.

Example :-

a) Confidentiality : (Service)

- Mechanism - Encryption, VPN tunneling, access control.

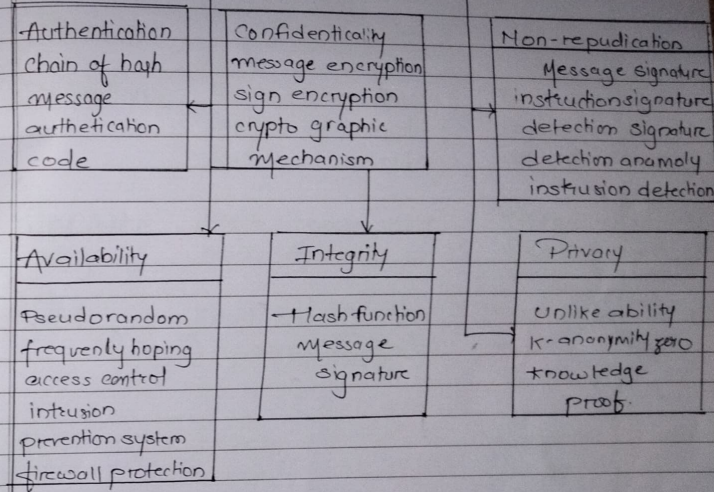
b) Integrity (Service):

- Mechanism - Hashing, Checksums.

c) Authentication (Service):

- Mechanisms - Password, Biometrics, Certificates.

Security Services and Security Mechanism



Explain Substitution & transposition techniques, Vigenere cipher, Playfair Cipher, Hill cipher with example.

If Substitution Techniques :-

- Substitution technique is one that the letter in the
- plaintext will be replaced by other letters/numbers/symbols.
- These plaintext units may be individual letters or character, letter pair, triplets or other combinations.
- Example : Caesar Cipher is an example of Substitution cipher.

plaintext : a b c d e f g h w x y z
ciphertext : e f g h i j k l z a b c

So,

plaintext : meet me after the party

ciphertext : phhw ph diwhu wkh sdwuo

II] Transposition Techniques :-

- Each character is positioned differently from its original position.
- Character position is changed but identity remains there.
- A letter near to original position get traced easily.
- Example : Rail-Fence Cipher is an example of transposition cipher.

plaintext : TRANSPPOSITION

keyword : 43152

4	3	1	5	2
N	A	T	S	R
I	S	P	T	O
X	N	I	X	O

cipher text : NIXASH TPISTXROO

III] Vignere Cipher :-

- Blaise de Vignere, he used a table known as the Vignere sequence to encipher message.
- In addition to the plaintext the Vignere cipher requires a keyword which is repeated so that the total length is equal to the length of the plaintext.
- The Vignere key strength does not depend on the plaintext character; it depends only on the position of the character in the plaintext.

- Example :

INPUT : PLAINTEXT : GEEKSFORGEES

Keyword : AYUSH

OUTPUT : Ciphertext : GCYCZFMLYLEIM

For generating key

The plain text

The keyword "AYUSH" generates the key "AYUSHAYUSHAYU"

IV] Playfair Cipher :-

- The Playfair cipher was the first practical substitution cipher.
- The technique encrypts a pair of letters instead of single letters as the simple substitution cipher.
- The Matrix contains the alphabet that acts as the key for encryption of the plaintext.

Vigenere Cipher

Plaintext : apple
Key : snow

(P) Plaintext : a p p l e
 value : 0 15 15 11 4
(K) Key : s n o w s
 value : 18 13 14 22 18
(P+K) mod 26 : 18 2 3 7 22
Ciphertext : s c d h w

Plaintext : wearediscovered
Key : deceptive

a	b	c	d	e
<u>0</u>	1	2	3	4
f	g	h	i	<u>0</u>
5	6	7	8	9
k	l	m	n	o
10	<u>11</u>	12	<u>13</u>	<u>14</u>
p	q	r	s	t
<u>15</u>	16	17	<u>18</u>	19
u	v	w	x	y
20	21	<u>22</u>	23	<u>24</u>
z				
25				

The Hill Cipher

Encrypt the message 'exam'
using Hill Cipher with Key $\begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$

$$\text{Plaintext (P)} = \begin{bmatrix} e \\ x \end{bmatrix} \begin{bmatrix} a \\ m \end{bmatrix} = \begin{bmatrix} 4 \\ 23 \end{bmatrix} \begin{bmatrix} 0 \\ 12 \end{bmatrix}$$

$$\text{Key (K)} = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$$

$$KP = \begin{bmatrix} 12 & 8 \\ 18 & 1 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$$

$$KP \bmod 26 = \begin{bmatrix} 24 \\ 25 \end{bmatrix} \begin{bmatrix} 22 \\ 6 \end{bmatrix} = \begin{bmatrix} y \\ z \end{bmatrix} \begin{bmatrix} w \\ g \end{bmatrix}$$

$$\text{Ciphertext} = \underline{y3wg}$$

$$\begin{bmatrix} b & c \\ d & f \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

$$C = KP \bmod 26$$

a	b	c	d	e
0	1	2	3	4
f	g	h	i	j
5	6	7	8	9
k	l	m	n	o
10	11	12	13	14
p	q	r	s	t
15	16	17	18	19
u	v	w	x	y
20	21	22	23	24
z				
25				

- Example :

Plain text : "instruments"
 after split : 'in' 'st' 'ru' 'me' 'nt' 'sz'
 Explanation : Pair cannot be made with letter

Plain text : "Hello"
 after split : 'he' 'lx' 'llo'
 explanation : Here x is the bogus letter.

Hill Cipher :-

- Hill cipher represent a polygraphic substitution cipher that follows a uniform substitution across multiple level of blocks.
- Hill cipher is based on a particular mathematical topic of linear algebra and the sophisticated use of matrices in general as well as Euler for modular arithmetic.

- Example :

Input : Plaintext : ACT
 Key : GYBNQEURP
 Output : ciphertext : POH

Input : Plaintext : GIG
 Key : HILLMAGIC
 Output : ciphertext : SWK

Difference between Active Attack & Passive Attack.

Parameter	Active Attack	Passive Attack
Definition	Active attack modification in information take place.	Passive attack modification in information does not take place.
Complexity	High	Low
Possibility	The prevention possibility of active attack is High.	The prevention possibility of passive attack is low.
Duration	The Duration is short.	The Duration is Long.
Purpose	The purpose of an attack the original information is modified.	In Passive attack original information is unaffected.
System	Cause system damage.	No system damage.

Explain in details Block cipher mode of operation.

- A Block cipher encrypts a fixed size block.
- To handle longer message mode of operation specify how blocks are linked together.
- Most modes use an initialization vector.

Block cipher

Stream Cipher

1) Encrypt a block of text at a time

1) Encrypt data one byte or bit at a time

2) Time Consuming

2) less consuming

3) Used in chain modes

does not use

4) Slow & easy to implement

HW & easy to implement

5) If one bit is lost, whole block is corrupted.

If one bit is lost, only bit is corrupted.

6) DES, AES, RSA

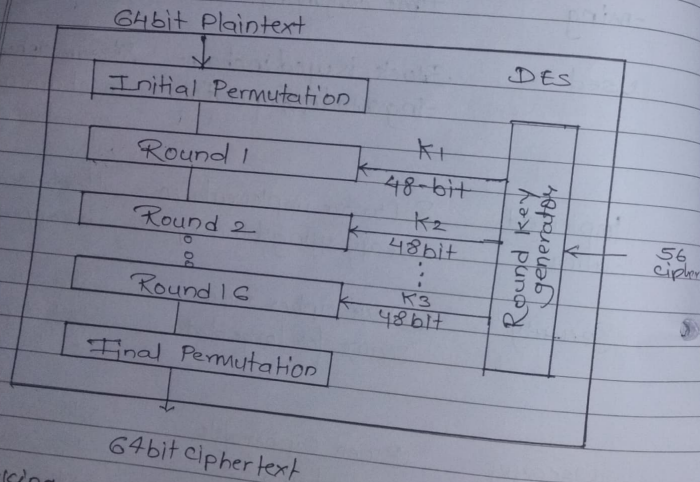
RCA

Q.6] Difference between Block Cipher & Stream Cipher.

<u>Parameter</u>	<u>Block Cipher</u>	<u>Stream Cipher</u>
Definition	Block cipher is main method of encrypting text in which key & algo.	In stream cipher keys and algorithm are applied.
Time Consuming	Block cipher is more time consuming.	Stream cipher less time consuming
Used	Block is used in chaining mode of operation.	Stream cipher doesn't used in chaining mode of operation.
Implementation	Software implementation is easy using block cipher	Hardware implemented is easy using stream.
Security	Block cipher more secure for large data.	Stream cipher secure but depends heavily on key stream.
Uses/cases	Block Cipher file encryption disk Encryption.	Stream cipher real time communication streaming.

Q.7 Explain Working of DES with block diagram.

- The Data Encryption standard algorithm is a symbol key block cipher.
- Triple DES is a symmetric key block cipher which applies the DES cipher in triplicate.
- At the encryption site, the algorithm takes the plain text in 64 bit blocks and converts them into 64 bits ciphertext using 56 bits keys.



Working
Input plaintext
The plaintext is divided into 64 bit blocks
each block is processed separately.

ii) Initial Permutation

- The 64bit plaintext block undergoes initial permutation
- This is a fixed rearrangement of bits.

iii) Key generation

- Original Key is 64 bit but 8 bit are parity bits.
- Effective Key size = 56 bits

iv) 16 Round of Encryption :

- DES uses 16 identical rounds

v) Final swap

- After the 16th round the left and right halves are swapped one last time.

vi) Final Permutation

- Output obtained is the ciphertext.

Q.8

Explain "Diffie-Hellman" key exchange algorithm with suitable eg.

- The "Diffie-Hellman" algorithm is a key exchange method used to securely share a secret key over an insecure communication channel.
- It does not encrypt message directly.
- It is used to generate a shared secret key.

Key :- The two users agree on some public value
exchange public key and independently compute

same secret.

- Algorithm

1) choose Public Value:-

Both user agree on:

- prime number P Ex:- $P=23$

- primitive root g Ex:- $g=5$.

2) choose Private key:-

- user A select private key a Ex:- $a=6$

- user B select private key b Ex:- $b=15$

These are kept secret

3) Generate Public key:-

- They exchange A and B openly Ex:- $a \rightarrow 8$

4) Compute shared Secret key

- $Key = B \bmod P$

- Both get the same value k which become the secret key

Ex:-

$$K = 19 \bmod 23 = 2$$

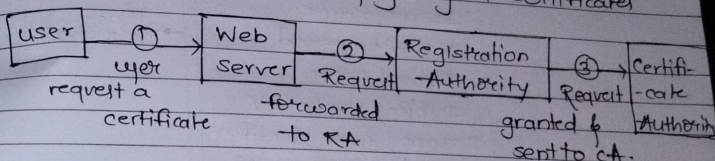
$$K = 8 \bmod 23 = 2$$

shared Secret Key = 2

Write a short note on X.509 digital certificate

X.509 digital certificate is a standard format used to verify the identity of an entity in a network. It is widely used in PKI.

- A public key is a large numeric value used to encrypt data or checks the legitimacy of a digital signature
- A PKI moreover is the underlying framework that enables entities like user and server to securely exchange information using digital certificates



Type:-

- Root Certificate
- Intermediate Certificate
- SSL Certificate
- Code Signing Certificate
- Email Certificate

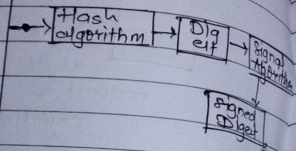
Advantage:-

- Strong Authentication
- Global Standard
- Support Confidentiality

Limitations:

- Certificate manage is complex
- Depend on trust in CA.
- Can be costly

version number	
serial number	
Signature algorithm ID	
Issure name	
Validity period	→ Hash algorithm → Digest → Digital Signature
Subject name	
Subject publickey	
Issure Unique Identifier	
subject Unique Identifier	
Extensions	
Signature	



where

- M = Plaintext
- C = ciphertext

Ex :-

let $m = 88$

$$C = 88^7 \bmod 187 = 11$$

$$\text{cipher text} = 11$$

Description :-

formula

$$M = c^d \bmod n$$

$$M = 11^{23} \bmod 187 = 88$$

To decrypt the cipher text c use the private key (n, d) and get the original data using the formula.

Encryption :- To encrypt a message M it is first converted to numerical representation using ASCII and other code encoding schema.

$C = m^e \bmod n$ where C is the cipher text and e and n are part of public key.

Q10] Explain RSA Algorithm with example.

→ Rivest Shamir Adleman algorithm is an example of an asymmetric or public key cryptography algorithm which means it works on 2 Key.

Key Generation :- Choose two numbers say p & q .

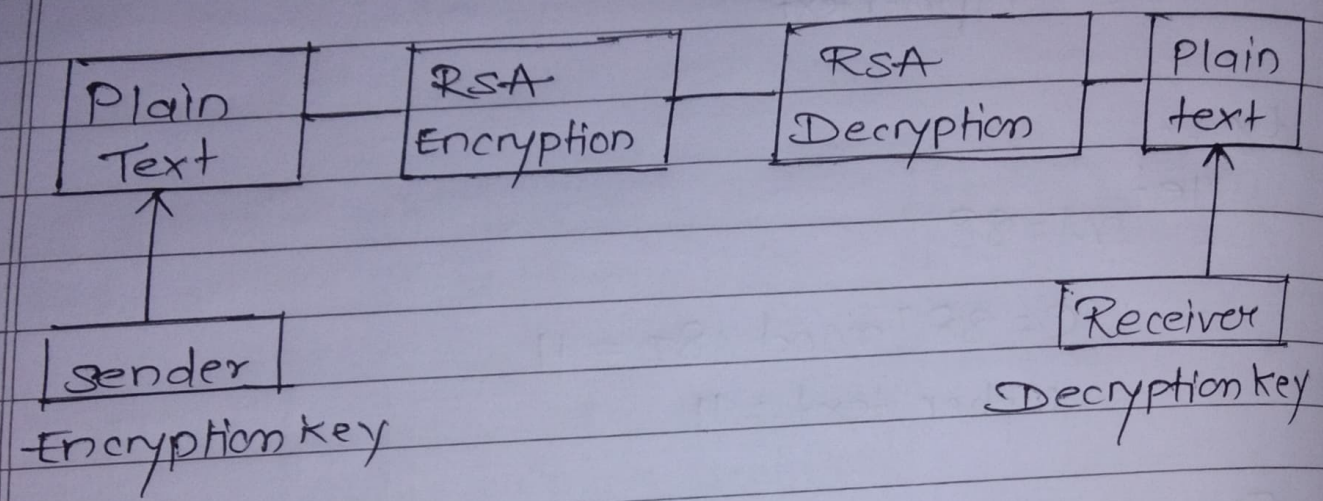
- Calculate the product of prime $n = p * q$.

Ex :-

- Euler Totient function $\phi(n)$ as

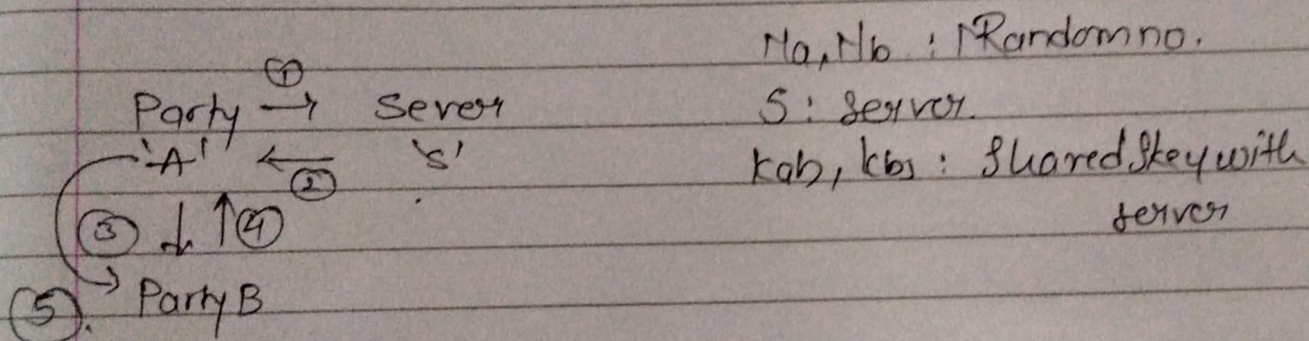
$$\phi(n) = \phi(p * q) = \phi(p) * \phi(q) = (p-1) * (q-1)$$

Communication channel



* Needham - Schroeder authentication protocol.

- Protocol used to securely distribute the session keys between two parties using a trusted Third party (KDC)
- solves problem of "How do we talk securely if we've never met?"
- key exchange protocol for authentication over an insecure network.



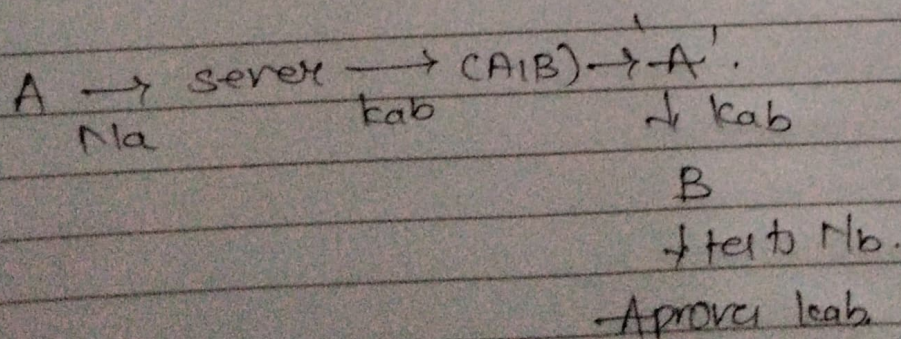
$A \rightarrow S$: A, B, N_a (A request to talk to B, with nonce N_a)

$S \rightarrow A$: $\downarrow$ Sends session k_{ab} , encrypted for A, B.

$A \rightarrow B$: (A forward k_{ab} to B)

$B \rightarrow A$: B tests k_{ab} with nonce N_b .

$A \rightarrow B$: A proves knowledge of k_{ab} .



--	--	--

Q5. Explain Public Key Distribution in detail.

Answer: Public Key Distribution is how we share public keys so anyone can send us encrypted messages. The main challenge is ensuring the key actually belongs to the person claiming it (avoiding fake keys).

Methods of Distribution:

1. **Public Announcement:** The user broadcasts their key (e.g., puts it on a website).
Weakness: Anyone can fake the announcement.
2. **Publicly Available Directory:** A trusted organization keeps a phonebook-like directory of names and keys.
3. **Public Key Authority:** A dynamic server that provides public keys upon request.
4. **Public Key Certificates (Best Method):** A Certificate Authority (CA) signs your public key. This is like a digital ID card that proves the key is yours. This is used in HTTPS websites.

* ~~X.509~~ X.509 Digital certificate.

- ① X.509 is the standard format of digital certificate.
- ② It is used to provide authentication and security in networks.
- ③ It binds the public key with the identity of user or server.
- ④ It is issued and digitally signed by trusted entity called a CA (certificate authority).
- ⑤ It is used in HTTPS ~~and~~, email security, VPN etc.

- main contents of X.509 certificate are as follows.

version			
serial number			
signature algorithm identifier			
issuer name			
validity period			
subject name			
public key information	✓		
issuer unique ID			
subject unique ID			
extension			

version			
serial number			
signature algorithm identifier			
issuer name	v1	v2	v3.
validity time			
subject name			
public key information	✓		
issuer unique ID			
subject unique ID		✓	
extensions			✓

working:-

- ① The user / server generates the public key.
- ② The public key is send to CA with identity proof.
- ③ CA verifies the identity and issues the X.509 certificate.
- ④ Then the certificate is digitally signed by CA.
- ⑤ while communication user verifies the CA's signature to trust the public key.